Kojo B. Osei-Ntansah

Aerospace Engineering PhD student at Georgia Tech studying autonomous aerial systems that maintain reliable flight and sensing under environmental uncertainty. Background spans experimental flight research, fault-tolerant control, scientific payload integration, and additive manufacturing, with first-author publications at AIAA SciTech 2025 and 2026.

FAA Part 107 Certified Remote Pilot

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EDUCATION

Georgia Institute of Technology Aug 2026 to Present

Ph.D. in Aerospace Engineering · Atlanta, GA · Advised by Dr. Jonathan Rogers, AREAL

Planned coursework: AE 6520 Advanced Flight Dynamics; AE 6530 Multivariable Linear Systems and Control; AE 6580 Aerospace Nonlinear Control; AE 6511 Optimal Guidance and Control; AE 6505 Random Processes and Kalman Filtering; AE 6513 Mathematical Principles of Planning and Decision-Making for Autonomy; AE 6333 Rotorcraft Design I; AE 6503 Helicopter Stability and Control; AE 6210 Advanced Dynamics I; ME 6407 Robotics; MATH 6643 Numerical Linear Algebra; MATH 6221 Probability Theory for Scientists and Engineers

Howard University Aug 2022 to May 2026

B.S. in Mechanical Engineering · Washington, DC · GPA 3.86 · Summa Cum Laude · Karsh STEM Scholar (full-ride merit scholarship)

PUBLICATIONS

- Osei-Ntansah, K., Reliford, A., and Smith, S. T., 'Development of Autonomous Strategies for High Resolution Data Acquisition for sUAS-Based Atmospheric Research,' AIAA SciTech Forum, 2026.
- Osei-Ntansah, K., Reliford, A., and Smith, S. T., 'Design and Implementation of 3D Printed Sensor Mounts for sUAS-Based Atmospheric Research,' AIAA SciTech Forum, 2025, p. 0192.
- Reliford, A., Smith, S. T., Osei-Ntansah, K., Stewart, A., and Jackson, D., 'From Calibration to Flight: Assessing the Alphasense OPC-N3 for Aerial Data Collection,' AIAA SciTech Forum, 2025, p. 1030.
- Reliford, A., Smith, S., Osei-Ntansah, K., Stewart, A., and Jackson, D., 'sUAS and AI/ML Technologies for High-Resolution Air Quality Monitoring in Urban and Industrial Regions,' AGU Fall Meeting Abstracts, Vol. 2024, No. 1995, A43D-1995, Dec. 2024.
- Osei-Ntansah, K., and Yu, J., 'Streamlining Processes of sUAS for Rapid Deployment,' 2026 DoW HBCU/MI Summer Research Internship Program Intern Symposium, DEVCOM Army Research Laboratory, Aberdeen Proving Ground, MD, 2026. (Distribution Statement A)

RESEARCH EXPERIENCE

Graduate Research Assistant

Aerial Robotics and Experimental Autonomy Lab (AREAL), Georgia Institute of Technology Aug 2026 to Present

- Extending an extended Kalman filter for range-only localization of radio-tagged birds from RSSI, using a log-distance path-loss measurement model built from measured sensor data rather than assumed propagation constants.
- Addressing the estimator limits that range-only geometry imposes: censored below-threshold measurements, the information carried by missed detections, and mirror ambiguity in the range-only solution.
- Investigating observability-aware flight planning to improve localization over standard lawnmower sweep patterns, evaluated through Monte Carlo trade studies on flight speed, tag burst interval, and detection range using behavior-based bird motion models.
- Characterizing receiver detection performance against range, antenna orientation, vehicle attitude, SDR gain, and environment through controlled flight and ground tests, and diagnosing unit-to-unit range variation across identical receivers by isolating vibration, gain control, and antenna-mounting effects.
- Developing the drone-mounted, Motus-compatible VHF receiver behind the measurement (SensorGnome, Raspberry Pi, SDR, GPS), evaluating RF front-end components (SDRs, LNAs, bandpass filters, whip and Yagi antennas) for signal-to-noise ratio, and integrating the payload with a Holybro X500 V2 airframe while balancing mass, endurance, antenna placement, and onboard RF noise.

Student Researcher

@FTERLab, Applied Fluids and Thermal Engineering Research Laboratory, Howard University Aug 2023 to Jul 2026

- Designed and 3D printed scientific sensor mounts in SolidWorks and Fusion 360 carrying a 753 g Alphasense OPC-N3 and iMet-XQ2 payload, balancing center of gravity, propeller and landing gear clearance, and access to sensor ports across repeated field deployments.
- Standardized the same payload across every quadrotor in the lab so measurements were platform agnostic, letting datasets be compared across airframes instead of being tied to a single vehicle.
- Ran CFD in ANSYS Fluent to position the sensor inlet approximately 0.2 m above the vehicle body across

3,000 to 7,000 RPM rotor speeds and 5 m/s headwind and 2 m/s crosswind cases, so rotor induced turbulence would not bias the particulate measurements the aircraft existed to collect.
- Validated the integrated system against stationary reference sensors in hover testing, demonstrating research grade measurements from rapidly replaceable printed hardware instead of costly custom machined payload structures.
- Developed Python and DroneKit mission software on a Raspberry Pi companion computer commanding a Cube Orange autopilot over MAVLink, automating four repeatable sampling geometries and removing the pilot to pilot variation that had made the lab's atmospheric measurements difficult to reproduce.
- Ran the Alphasense OPC-N3 as a supervised subprocess with a file handshake start and stop, confirming logging had initialized before the vehicle proceeded and closing files cleanly on abort, eliminating the corrupted datasets that had previously cost the team entire flights.
- Executed 50+ flight tests over three years across five quadrotor platforms, advancing each capability through a staged risk reduction process from SITL simulation and propellers off bench checks to full autonomous sampling flights.
- Diagnosed field failures from flight logs, including GNSS and EKF degradation traced to companion computer electromagnetic interference, yaw imbalance, and potential thrust loss warnings, isolating power, calibration, and propulsion causes to return aircraft to service between test campaigns.
- Assembled, wired, and commissioned research quadrotors around Cube Orange autopilots and Raspberry Pi companion computers with GNSS, RFD900x and Herelink telemetry, and scientific payload provisions, working through a staged safety and commissioning checklist from continuity and polarity checks to sensor calibration, failsafe and geofence configuration, and incremental autonomous flight.
- Expanded the lab's flight ready fleet, adding test capacity and letting campaigns continue when a single airframe was down for maintenance.
- Integrated the aircraft's communication paths into one field workflow: Raspberry Pi to Cube over UART/USB MAVLink, RFD900x on the TELEM port, Herelink for pilot control and video, a Pi Wi-Fi access point for field SSH without internet, and Arducam streaming to a ground computer — managing baud rates, network IDs, Linux permissions, bandwidth competition, and link-loss behavior so loss of the Pi network never removes RC control.
- Built flightgen, a Flask web application deployed as a systemd service on the aircraft's Raspberry Pi and reached over the drone's own Wi-Fi network, letting an operator generate, launch, monitor, and abort an autonomous mission from a browser in the field without touching a terminal.
- Generated standalone DroneKit mission scripts from pattern parameters for hover, box, lawnmower, and polygon geometries, converting local north and east offsets to geographic waypoints and ending each leg on an arrival distance check rather than a fixed timer.
- Implemented run control that launches each mission in its own process group so a single abort terminates the whole process family, including the particulate sensor's child process, with a background thread streaming live mission output to the browser and a lock preventing two operators from starting a flight at once.
- Wrote a separate abort path that commands LAND or RTL and confirms the flight controller actually changed mode, keeping emergency behavior on the autopilot rather than depending on the companion computer staying alive.
- Created a data workflow aligning optical-particle-counter samples with autonomous flight records — corrupt-row filtering, timestamp normalization, flight-phase segmentation, and cautious resampling — applied to June-July 2026 datasets comparing extended vs. unmodified inlet configurations, airborne vs. stationary behavior, 15 m and 20 m sampling altitudes, and controlled smoke-response scenarios.

Research Intern, DoW HBCU/MI Summer Research Internship Program

DEVCOM Army Research Laboratory, Manufacturing Science and Technology Branch — Jun 2026 to Aug 2026

- Redesigned printable structural components of a fully 3D-printable sUAS quadcopter in Fusion 360, eliminating screws and fasteners to cut frame weight 52% (600 g to 290 g) while removing the assembly steps that required specialized knowledge, so non-specialist personnel could build the aircraft correctly under time pressure.
- Authored assembly guides and configuration checklists validated through repeated build exercises measuring assembly time and system readiness; delivered to a U.S. Army infantry division of 10,000+ soldiers, cutting a build from a full workday of hunting scattered online instructions to a repeatable 2-3 hour protocol.
- Designed and fabricated a mounting solution integrating the Avatar HD Kit V2 FPV camera while preserving field of view and airframe compatibility, extending the platform's reconnaissance capability without a redesign of the vehicle.
- Presented the work at the 2026 DoW HBCU/MI SRIP Intern Symposium (public release, Distribution Statement A).
- CAD modeled and 3D printed a fully printable modular 5-inch quadcopter airframe in Fusion 360 using snap-fit joints, so arms and structural components can be swapped in the field without fasteners, tools, or adhesives - the failure mode that normally grounds a printed aircraft after a crash.
- Sized arm cross-sections against motor thrust and landing loads while accounting for printed-layer anisotropy, print orientation, motor fastener placement, and propeller clearance, keeping the airframe printable without

support material.

SURE Research Intern

Aerial Robotics and Experimental Autonomy Lab (AREAL), Georgia Institute of Technology May 2024 to Aug 2024

- Supported a Kalman filter based fault estimation effort for fixed wing elevon failure in MATLAB and Simulink, working through how comparing expected against observed aircraft response lets a degraded control surface be detected and its authority reallocated in flight.
- Helped prepare the test plan for hardware in the loop simulation, defining the setup needed to expose the timing and interface behavior that pure software simulation cannot show.
- Developed and iterated a rudder redesign through 3 CAD revisions, evaluating control surface area, moment arm, hinge placement, and servo linkage geometry to recover control authority lost to the failed elevon.

Research Intern, DoD HBCU/MI Summer Research Internship Program

DEVCOM Army Research Laboratory May 2025 to Aug 2025

- Designed and prototyped a two-axis 3D-printed gimbal in Fusion 360 for an Arducam optical sensor, reaching roughly 90 degrees of useful travel per axis while aligning camera mass near each rotation axis to reduce required servo torque and payload mass on the aircraft.
- Applied structural analysis and vibration-isolation principles to optical sensor mounting for fixed-wing flight, trading printed wall thickness against inertia and managing wire clearance across the full motion envelope so pointing would hold under flight loads.
- Worked within defense documentation and design-review practice, producing a public-release poster documenting the prototype and its mechanical and flight validation plan.

NSF REU Research Intern

Unmanned Aerial Systems Laboratory, Cal Poly Pomona May 2023 to Aug 2023

- Developed tandem-wing eVTOL concept geometry in OpenVSP and detailed components in SolidWorks and Fusion 360, evaluating wing loading, center-of-gravity envelope, and propulsion placement against conflicting hover and cruise requirements.
- Fabricated and assembled prototype structures and integrated motors, servos, control hardware, wiring, and batteries, including machine-shop work on airframe components.
- Conducted X-Plane and autopilot simulation followed by ground and flight testing, revising the configuration in response to observed stability and control behavior and supporting propulsion performance optimization for the lab's NSF REU program.

PROJECTS

Developer

Personal project Aug 2026 to Present

- In-progress tool that ingests ArduPilot BIN/CSV flight logs and reports vibration levels, EKF innovations, GPS quality, battery and current draw, motor-output balance, commanded vs. actual attitude, failsafe events, and waypoint error — with automatic anomaly flagging. Built against a real corpus: the GNSS glitches, EKF resets, yaw imbalance, and thrust-loss warnings from three years of my own flight testing serve as the labeled cases.
- In-progress onboard perception project: detection and tracking running on a Raspberry Pi with an Arducam feed, closing the loop by commanding gimbal or vehicle yaw to hold a target centered. The engineering questions are the embedded ones — inference rate against frame rate, end-to-end latency budget, and what pointing error the loop can actually hold once the airframe is moving.
- In-progress GPS-denied navigation testbed: camera calibration, then AprilTag and ArUco pose estimation, transformed into the vehicle body/NED frame, timestamped, and sent to a Cube Orange as MAVLink vision-position messages with EKF source selection configured to accept them. Later phases add optical-flow odometry with rangefinder scale and drift evaluation over closed paths. Validated on a staged ladder from bench markers to restrained motion to flight.
- In-progress multi-vehicle architecture: each aircraft keeps an independent flight controller and safety pilot, a companion computer runs vehicle-level logic, and formation positions are defined as offsets from a leader or virtual leader, with a global enable that prevents accidental activation. Proven in SITL across staged steps — simulated pairs, then one real vehicle following a virtual leader — studying how communication latency and vehicle dropout degrade formation error before any two-aircraft flight.
- Building a ROS based robotics project with Arduino microcontrollers handling low level sensing and actuation, to work fluently in ROS nodes and topics and in the C++ side of the robotics stack.

Senior Project Team Lead

Howard University — Aug 2025 to May 2026

- Designed the test fixture and specimen geometry (69.85 x 7.62 mm) in CAD and planned the fatigue campaign using Basquin relationships, predicting 10^6 to 10^7 cycle lives at 1,250 to 1,750 N loads.

HONORS AND AWARDS

- GEM PhD Fellowship Awardee (declined), The National GEM Consortium, 2026
- B.S. Mechanical Engineering, Summa Cum Laude, Howard University, 2026
- Karsh STEM Scholar (full-ride merit scholarship), Howard University, 2022
- FOCUS Scholar (2024 and 2025), Georgia Institute of Technology, 2025
- Certificate of Completion, Aerospace Propulsion Course, Massachusetts Institute of Technology, 2025

TECHNICAL SKILLS

Design & Analysis: SolidWorks, Fusion 360, Creo PTC, OpenVSP, ANSYS, FEA, CFD, Structural analysis

Software: Python, MATLAB, Simulink, DroneKit, MAVLink/pymavlink, Flask, HTML/CSS, Arduino, Raspberry Pi, Linux, Git, Monte Carlo simulation

Estimation & Autonomy: Extended Kalman filtering, Particle filters, Range-only localization, Path-loss modeling, Information-driven path planning, Sensor fusion

RF & Sensing: Software-defined radio (Airspy, RTL-SDR), VHF antennas, LNAs and bandpass filters, SNR and link-budget analysis, GPS integration, SensorGnome/Motus pipeline

Flight Systems: ArduPilot, PX4, Cube Orange, Mission Planner, QGroundControl, SITL/HITL simulation, X-Plane, RFD900x/Herelink telemetry

Fabrication & Test: FFF 3D printing, CNC machining, Electronics integration and wiring, Thrust testing, Flight testing (50+ tests across 5 quadrotor platforms)

SERVICE

- **Team Captain, Men's Lacrosse**, Howard University (2023 to 2026)
- **Executive Board, American Society of Mechanical Engineers**, Howard University (2022 to 2023)
- **Member, National Society of Black Engineers**, Howard University (2022 to 2026)
- **Member, Robotics Club**, Howard University (2022 to 2026)